Clean Energy and Water (CEW) Laboratory

RIT
Kate Gleason College of Engineering

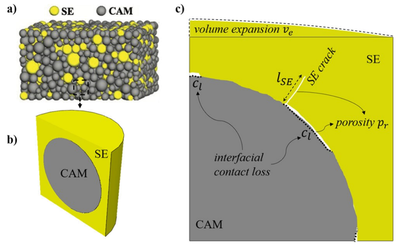
Dr. Howard Tu

Overview

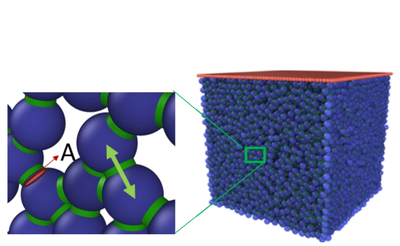
CEW Lab's mission is to push the development of sustainable energy and environment, with the data-calculation-experimental approach. The scientific target is to understand the coupling between electro-chemo-mechanics, electrochemical ion insertions, and their application in energy storage and water purification.

Theoretical Modeling

Computations (density functional theory, molecular dynamics) & simulations (finite element, phase field) for fundamental understandings.



Schematic of cathode in solid battery.



Schematic of solid electrolyte

Butler-Volmer reactions:

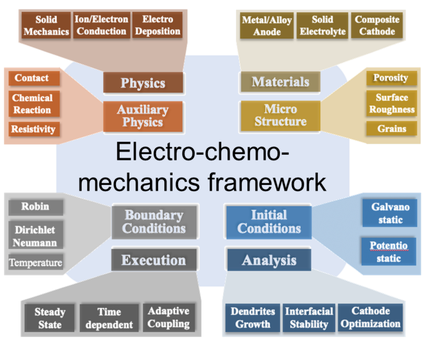
$$i_n^k = i_{exc}^k \left(e^{\frac{\alpha_a F \eta_k}{RT}} - e^{-\frac{\alpha_c F \eta_k}{RT}} \right)$$

Diffusion in CAM:

$$\frac{\partial c_{Li}}{\partial t} = \nabla \cdot (M_{Li} \nabla \mu_{Li})$$

Migration:

$$\boldsymbol{i}_{M^+} = -\frac{\sigma_{M^+}}{F} \nabla \tilde{\mu}_{M^+}$$



Phase Segregation:

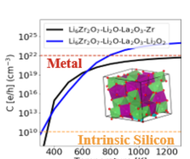
$$\mu_{Li} = \mu_{Li}^{eqm} - \nabla \cdot \left(\frac{\kappa}{\rho^2} \nabla c_{Li} \right)$$

Electronic Conduction:

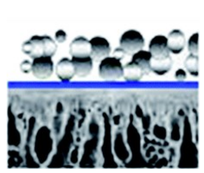
$$\boldsymbol{i}_{e^-} = \frac{\sigma_{e^-}}{F} \nabla \tilde{\mu}_{e^-}$$

Fracture in material:

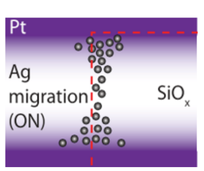
$$J := \int_\Gamma \left(W \, \mathrm{d}x_2 - \mathbf{t} \cdot \frac{\partial \mathbf{u}}{\partial x_1} \, \mathrm{d}s \right)$$



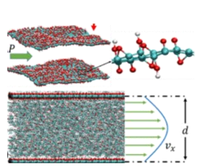
Ion/electron conduction with DFT calculations



Membrane fouling in CDI



Uncontrolled filament in ECRAM



Ion/water transport with Molecular Dynamics

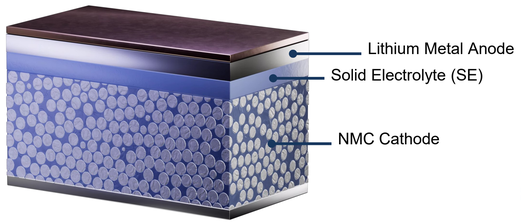
Energy Storage

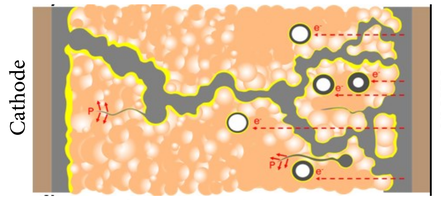
Energy Storage

Solid-state batteries (SSBs) represent a groundbreaking advancement in energy storage technology, utilizing solid electrolytes instead of liquid ones, offering advantages such as enhanced safety, higher energy density, and faster charging capabilities.

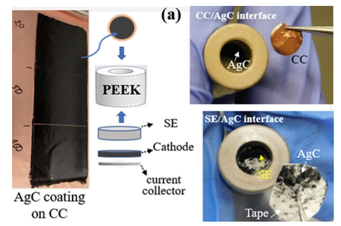
Projects:
- Dendrite Suppression
- Anode-free Architecture
- Battery Mechanics

Schematic of a solid-state battery

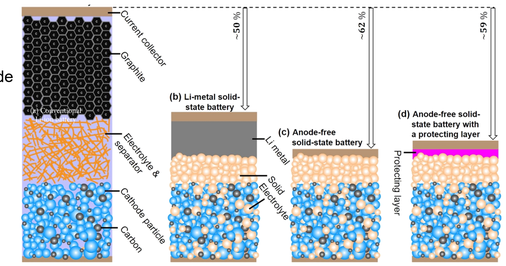




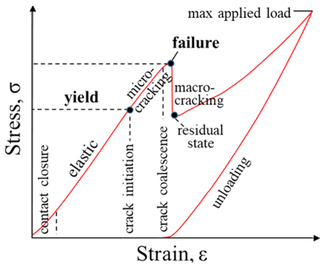
Schematic of Li dendrite solid-electrolyte penetration during battery cycling. Dendrites cause SSB short circuit.



Schematic and photos of anode-free SSB.



Roadmap for next-generation batteries.



Stress-strain curve of SE.

Water Purification

Capacitive Deionization (CDI) is an innovative water desalination method using charged electrodes, known for its energy efficiency, cost-effectiveness, and environmental friendliness.

Projects:
- Hazardous ions removal
- Valuable ions extraction
- Water desalination

CDI system:

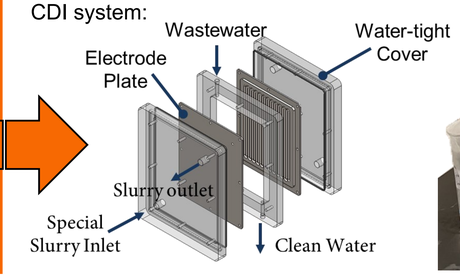


Photo of real CDI system



Combining the CDI and SSB technology for energy generation and lithium capture, with the following patented hybrid CDI-SSB device:

